QP CODE: 24900032

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCBOT100, MG1DSCBBT100 - FASCINATING WORLD OF PLANT SCIENCES

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answers

Answer 10 questions. each question carries 1 mark.

1. **Give two examples of fibre yielding plants.**
   Coconut, Jute, Cotton, Hemp etc ….or any other fibre yielding plant
2. **Which book by Carl Linnaeus introduced the binomial nomenclature system?**
   Species Plantarum
3. **Give the minimum contents in a field book.**
   Collection no., name of the plant, family, date and place of collection…etc (**Give full marks to any two valid points**
4. **Write two examples of algae.**
   Sargassum, Nostoc, Anabaena, Cladophora …etc (Liberal Valuation)
5. **Name one edible mushroom species.**
   Oyster Mushroom, Button mushroom, *Agaricus* etc (Liberal valuation)
6. **What is the dominant generation in the life cycle of gymnosperms?**
   Sprorophyte
7. **Give two examples for biomimicry.**
   Lotus leaf effect, *Citrus maxima* fruit wall…..etc
8. **Which group of land plants have gametophytic plant body as dominant phase in their life cycle?**
   Bryophyte
9. **What is scanning electron microscopy?**
   Advanced microscope used for getting high magnification surface images
10. **How does the total magnification of a compound microscope get calculated?**
    Multiply the resolution power of objective and eye piece (Liberal valuation)
11. **Name one application of centrifugation.**
    Separation of components from various crude samples
12. **What is chromatography?**
    Analytical technique used for separation purification and identification of different chemical constituents

(10 × 1 = 10)

Part B

Short Essay Type Questions

Answer 6 questions. Each question carries 5 marks.

13. **Briefly analyse the evolutionary trends in land plants.**

a. In the evolution of land plants the significant events include the change in body plans.
b. It included the emergence of multi-cellular organisms from unicellular ones.
c. The emergence of diploid sporophytes in advanced forms from haploid gametophytes.
d. Formation root, stem and leaf from simpler thallus forms.
e. The formation of flower and seeds in most advanced groups.

(Any other relevant points – give marks)

14. **Analyse the significance of fig leaves in lemur diets.**

- The behavior of self-medication by non-human animals by ingesting or topically applying plants, soils, etc. is called zoopharmacognosy.
- Pregnant lemurs in Madagascar eat fig leaves.
- They eat this for mainly three reasons –
  a. Inducing milk production,
  b. Killing parasites and
  c. Increase the chances of successful birth.
- Fig (high fiber content) - helps in digestion (antifatigue).
- Rich in nutrients, iron, potassium, calcium, vitamins A, B, and C, and protein

(Any five relevant points, give full marks)

15. **What are the advantages and limitations of free hand sectioning?**

Freehand sectioning is a technique commonly used to prepare thin sections of plant tissue for microscopic examination. The method consists of firmly holding a piece of fresh or preserved material between the left hand thumb and forefinger and cutting with a sharp razor.

Advantages

- Quick and cost-effective.
- Can be done almost anywhere, making it ideal for fieldwork or classroom demonstrations.
- Suitable for fresh and soft tissues.
- Limitations
- Less precise compared to microtome sectioning.
- Uniformity and thickness of sections can vary.
- Not ideal for very hard or tiny specimens.

(Give marks for any other relevant points)

16. **Briefly describe Agarose gel electrophoresis**.

- Agarose gel electrophoresis is a laboratory technique to separate, identify, and analyze DNA, RNA based on their size and charge in a porous agarose gel matrix.
- It is also used for the separation of proteins
- The movement of charged molecules in an electric field.
- DNA and RNA are negatively charged due to their phosphate backbone, so they migrate towards the positive electrode (anode) when an electric field is applied.
- Molecules are separated by size—smaller fragments move faster and travel farther through the gel, while larger fragments move slower.

(Any five relevant points)

17. What is DNA barcode? What is its significance in species identification?

DNA barcoding is a technique to identify and authenticate species in biological samples by analysing small, standardised DNA fragments known as DNA barcodes.

Significance

- DNA barcoding allows for the **quick identification** of species, even from small or degraded samples.
- DNA barcoding identifies species based on **genetic differences**, making it more accurate and independent of external factors.
- DNA barcoding can reveal **cryptic species** that look similar but are genetically distinct.
- Conservation of biodiversity by identifying and monitoring **endangered or endemic** species.
- Used in wildlife crime investigations to identify poached animals or smuggled plant products.

(**Definition – 2 marks, Significance 3 marks – any three relevant points)**

18. **Explain the significance of ethnobotany in pharmaceutical industry.**

- Ethnobotany is the 'science of people's interactions with plants' (any relevant definition).
- This includes the study of plants that have therapeutic applications.
- By studying the traditional practices, scientists have been able to extract the active ingredients for application in modern medicines.
- Traditional use of medicinal plants has proved to be an invaluable guide in the present day screening of drugs.
- Many modern medicines including reserpine, ephedrine, atropine, etc. have been discovered by following leads from the traditional use.

(Any other relevant points – give marks; Liberal valuation)

19. **Describe the principles and process of paper chromatography**.

- Paper chromatography is an analytical technique which separates substances based on their polarity and affinity for different phases
- Generally used for separating dissolved chemical substances by virtue of their different rates of migration across sheets of paper.
- The separation occurs based **on the differential affinity of substances for** the stationary phase (the paper) and the mobile phase (the solvent).
- When a mixture is applied to the paper and a solvent is allowed to move through it.
- The different components of the mixture move at different rates depending on their solubility and interaction with the paper and solvent.

(Any five relevant points)

20**. Why researches in crop improvement field considered as very crucial for human life?**

Crop improvement is the development of new crop varieties possessing high yield and disease resistance.This field is crucial for

a. Addressing the global food security challenges.
b. Promoting sustainable agriculture practices.
c. Development of crops those are resilient to climate change.

Using AI technologies, prediction of effects of genetic variation on crop varieties became possible.

This accelerates the breeding strategies and speed breeding accelerates the development of new crops.

(Give marks for relevant points)

(6 × 5 = 30)

Part C

Essay Type Questions

Answer any one question. question carries 10 marks

21. Compare artificial, natural and phylogenetic system of classification of plants

Artificial Classification	**Natural Classification**	**Phylogenetic Classification**
Only a few characters of the plants are being considered. Grouping of plants into herbs, shrubs and trees. Based on number of stamens and styles Eg. Classifications proposed by Theophrastus, Linnaeus	All the important characters of the plants are being considered, and classified according to their related affinities. Cytological, biochemical, developmental, physiological, anatomical, reproductive etc., were being considered. This system reflects the situation as it is thought to exist under natural conditions Eg. Bentham and Hooker classified plants, based on morphology, ultrastructure, anatomy, embryology and phytochemistry	The plants are classified according to their evolutionary and genetic affinities. However, it's difficult to classify the plants perfectly on the basis of evolutionary tendencies due to imperfection of fossil records. Eg. The systems of classification proposed by Engler in 1886.

(Valid and short descriptions, give full marks. Liberal valuation)

22. **Discuss psychoactive plants with relevant examples and analyze their significance in traditional healthcare systems**

They are plants which manufacture psychoactive chemicals that exert psychoactive effects. They are classified as stimulants, depressants, or hallucinogens.

Eg. Marula plant

- The Marula tree has deep spiritual significance in many African cultures, and is often associated with fertility, healing, and community.
- In many African traditions, this tree is also associated with femininity and motherhood. It is believed that women who are struggling with fertility can increase their chances of conceiving by drinking a tonic made from the bark of the Marula tree.
- The Marula tree is dioecious. This led to the traditional belief that drinking an infusion made from the bark of a male tree would lead to the birth of a boy, while drinking a bark infusion from the female tree would result in the birth of a girl.
- The tree is often associated with nurturing and caring, and is seen as a symbol of the interconnectedness of all living things.
- High vitamin C content of the Marula fruit- Elephant bulls feed almost exclusively on Marula fruits to boost their immune systems, prior to entering a period of musth.

(Valid and short descriptions…..give full marks)

(1 × 10 = 10)